

Sherry Wilson

Senior Python & Django Backend Developer

Colchester, England, UK | +44 7778 214285 | sherrywilly521@gmail.com |

linkedin.com/in/sherry-wilson-python-dev | Right to Work in UK (Graduate Route Visa – 2027)

PROFESSIONAL SUMMARY

Senior Python & Django Backend Developer with 3+ years of hands-on experience designing and delivering high-availability production APIs, background task systems, and cloud infrastructure on AWS. Proven track record of leading engineering teams, driving revenue growth, and improving system performance at scale. MSc Artificial Intelligence, University of Essex. Full right to work in the UK — no sponsorship required.

CORE TECHNICAL SKILLS

Backend Frameworks: Django, Django REST Framework (DRF), FastAPI, Flask, Celery, Flower

Databases & Caching: PostgreSQL, Redis, ChromaDB, SQL Optimisation

Cloud & DevOps: AWS EC2, Docker, Nginx, Load Balancers, Kafka, Microservices

AI / ML: LangChain, RAG Architecture, Gemini LLM, LLM Integration, Pandas, Matplotlib

API & Integration: REST API Design, GraphQL, HLS Video Streaming (FFmpeg), Background Processing

Other: Team Leadership, Agile, Git, Karate (API Testing)

PROFESSIONAL EXPERIENCE

Senior Python Backend Developer

Deepsense Digital Solutions — Chennai, India | Aug 2021 – Jul 2024 (3 years)

- Led a team of 3 backend developers delivering production systems for ITC (Classmate), Fabelle, Greencell Mobility (Neugo), and Nippon Paints.
- Designed and built scalable backend architectures with Django and Django REST Framework, serving thousands of concurrent users across multiple client platforms.
- Built and optimised GraphQL endpoints, reducing frontend-server data load by 30%.
- Implemented asynchronous background processing with Celery, Redis, and Flower, boosting system throughput by 30% and improving observability.
- Tuned PostgreSQL databases and crafted complex SQL queries for stable performance under peak loads.
- Deployed Nginx reverse-proxy and load-balanced infrastructure on AWS EC2 using Docker, ensuring high availability and security compliance.
- Integrated Kafka-based microservices communication, reducing system latency by 25%.
- Implemented HLS video streaming via FFmpeg for eLearning platforms, enabling efficient media delivery at scale.
- Integrated LangChain LLM features into production applications across client projects.
- Contributed to a 15% revenue uplift through scalable backend architecture for high-profile enterprise clients.

Python LLM Developer

AquSag Technologies — India | Sep 2025 – Oct 2025 (2 months)

- Analysed and triaged GitHub issues across trending open-source libraries to identify high-impact challenges.

- Evaluated unit test coverage and code quality for LLM performance assessment in real-world bug-fixing scenarios.
- Collaborated with researchers to design repositories and issues that effectively test LLM capabilities.
- Led and mentored junior engineers on technical projects.

AI Intern

Victoria Solutions — London Area, UK | Jul 2025 – Aug 2025 (2 months)

- End-to-end data analysis and cleaning with Pandas for AI model training datasets.
- Built data visualisation dashboards with Matplotlib to communicate insights to stakeholders.
- Preprocessed large datasets for ML pipelines, improving data quality and model performance.

Python Developer (Freelance)

Self-employed — Kerala, India | Jun 2020 – Jul 2021 (1 year 2 months)

- Delivered custom Django, Flask, and FastAPI backend solutions for diverse client projects with high security and scalability.
- Built and automated background job queues with Celery and Redis to streamline client workflows.
- Containerized and deployed scalable services using Docker across multi-environment setups.
- Integrated ML models into production environments with secure REST APIs.

Team Leader

Welshwood Manor — Colchester, UK | Jan 2026 – Present

- Supporting residents in a care setting while actively developing AI and backend projects during career transition into the UK tech industry.

KEY PROJECTS

Document Assistant API | *FastAPI + LangChain + ChromaDB + Gemini*

RAG-powered document Q&A; system that ingests PDFs and answers questions with accurate citations. Uses chunking, vector embeddings, and streaming responses via Gemini LLM.

HLS Video Streaming Platform | *Django + Celery + FFmpeg + HLS + AWS S3*

eLearning platform backend with adaptive HLS video streaming. Videos transcoded via FFmpeg into multiple quality tiers served through CDN-backed AWS S3. Achieved 30% faster media delivery at scale.

Multi-Tenant E-Commerce Backend | *Django DRF + GraphQL + Celery + Redis + Kafka + PostgreSQL + Docker*

High-availability REST and GraphQL backend for a multi-client retail platform. Features async order processing, real-time inventory updates via Redis pub/sub, and Kafka-driven microservices. Serving thousands of concurrent users.

ML Data Pipeline & Dashboard | *Python + Pandas + Matplotlib + FastAPI*

End-to-end data pipeline for AI model training — automated ingestion, cleaning, feature engineering with Pandas, and interactive visualisation dashboards built with Matplotlib for stakeholder reporting.

EDUCATION

Master of Science – Artificial Intelligence & Its Applications

University of Essex — Jan 2024 – Jan 2025

Bachelor of Computer Applications – Computer Programming & Specific Applications

University of Kerala — 2018 – 2021

CERTIFICATIONS

- FastAPI — The Complete Course 2025 (Beginner + Advanced)
- Advanced Django — Advanced Techniques in Django Framework
- UK Full Driving Licence